

EfficientNet-B2 Forest / Non-Forest Report

Model: EfficientNet-B2

Image Size: 260x260

Loss: Focal Loss (gamma=2.0)

Checkpoint Path: /kaggle/working/b2_best_model.pth

Dataset Split:

Train size: 24274

Validation size: 5202

Test size: 5202

Total size: 34678

Class Mapping:

forest = 0

non_forest = 1

Loaded Checkpoint:

Checkpoint epoch: 29

Best Validation Forest F1: 0.9898

TEST RESULTS:

Test Loss: 0.0042

Test Accuracy: 0.9973

Test Macro F1: 0.9954

Forest Class:

Precision: 0.9862

Recall: 0.9989

F1-score: 0.9925

Non-Forest Class:

Precision: 0.9998

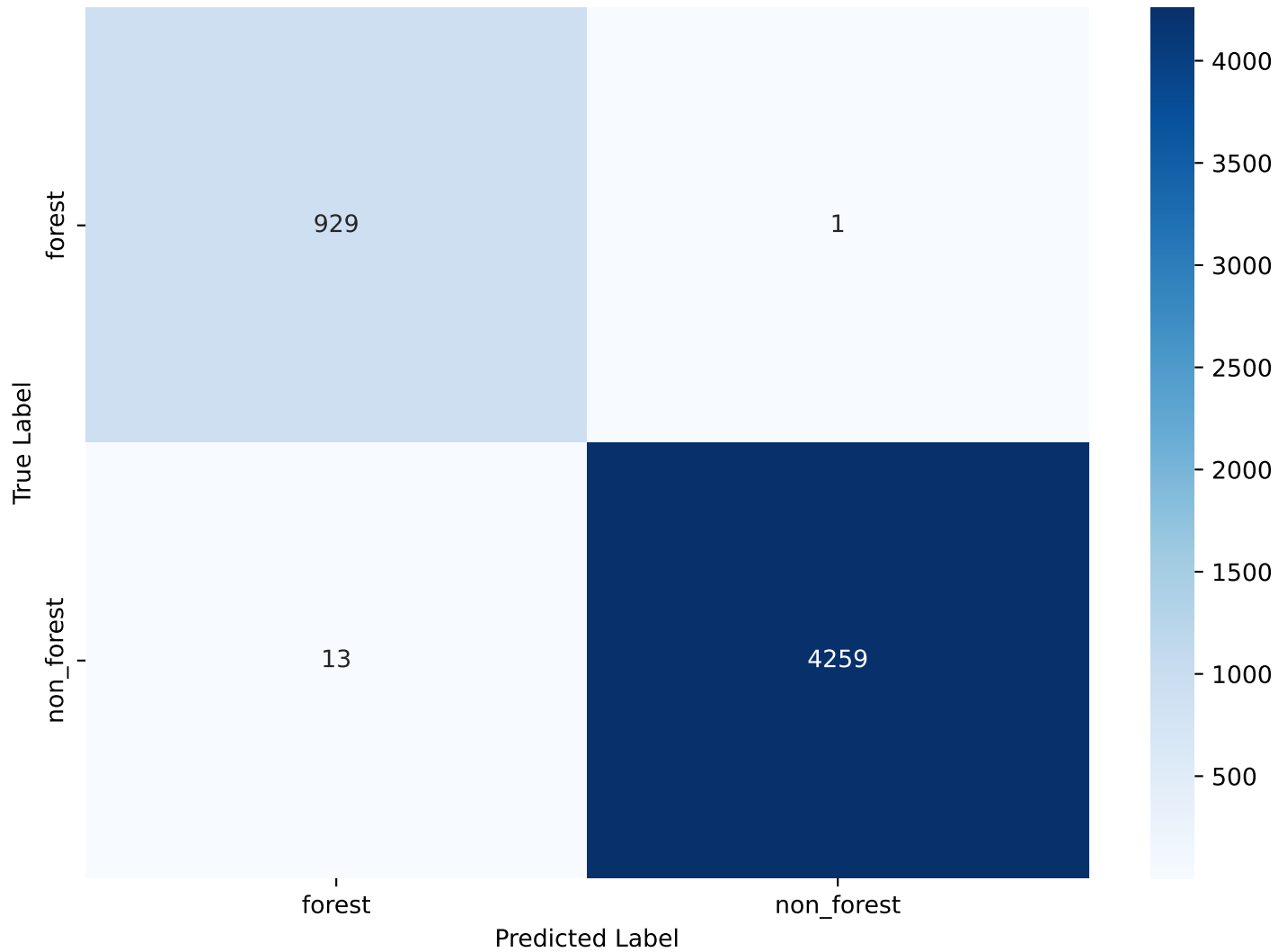
Recall: 0.9970

F1-score: 0.9984

Classification Report - Test Set

	precision	recall	f1-score	support
forest	0.9862	0.9989	0.9925	930
non_forest	0.9998	0.9970	0.9984	4272
accuracy			0.9973	5202
macro avg	0.9930	0.9979	0.9954	5202
weighted avg	0.9973	0.9973	0.9973	5202

Test Confusion Matrix



Model Training Report

Model: EfficientNet-B2

Image Size: 260x260

Batch Size: 32

Optimizer: AdamW

Loss: Focal Loss

Primary Metric: Validation Forest F1-score

Best Validation Result:

Stage: stage2

Epoch: 29

Val Forest F1: 0.9898

Val Macro F1: 0.9938

Val Accuracy: 0.9963

Val Loss: 0.0034

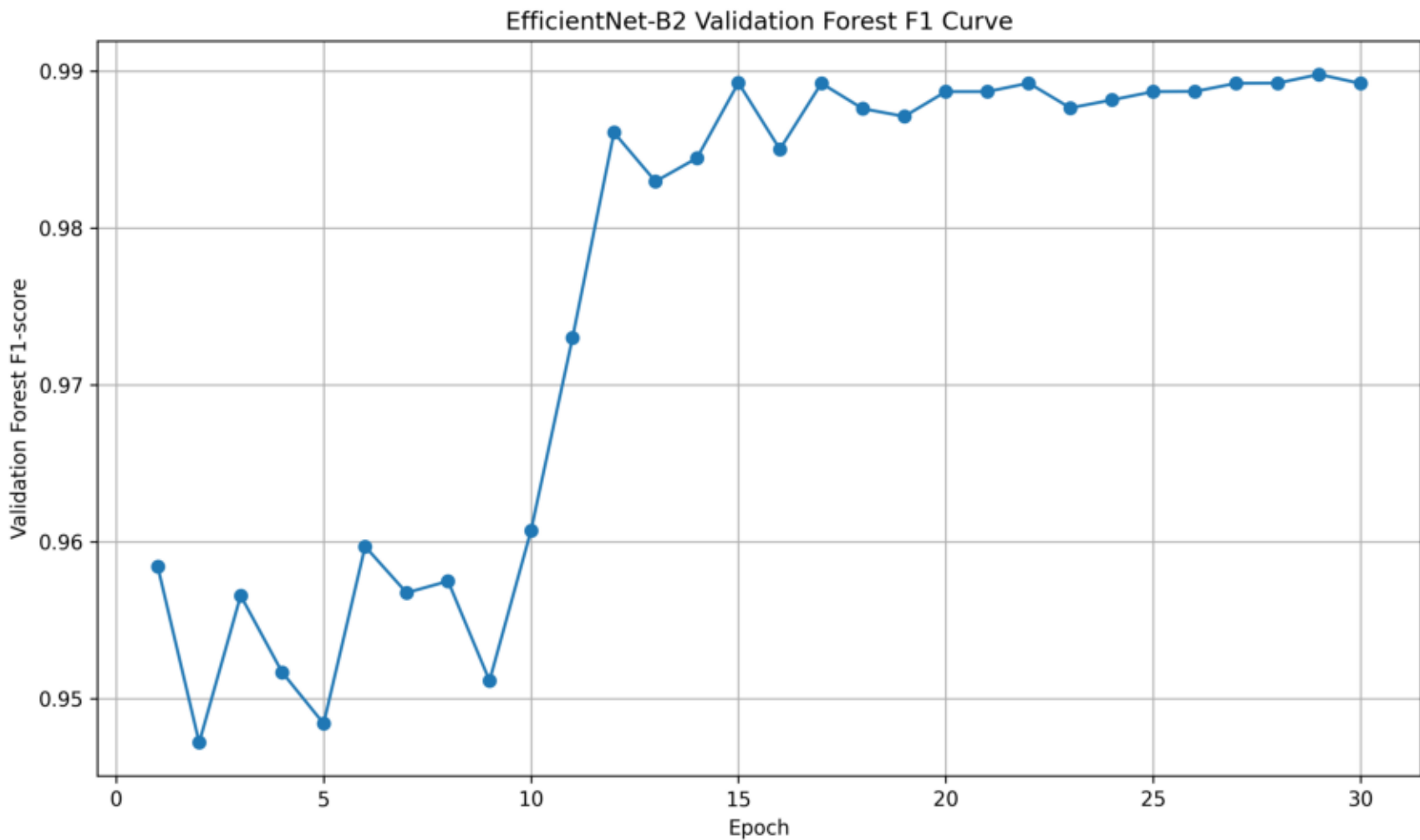
Best model path:

/kaggle/working/b2_best_model.pth

Epoch Results

stage	epoch	train_loss	val_loss	val_accuracy	val_macro_f1	val_forest_f1	time_sec
stage1	1	0.0355	0.0134	0.9850	0.9746	0.9584	463.2747
stage1	2	0.0278	0.0127	0.9810	0.9678	0.9472	431.0470
stage1	3	0.0296	0.0113	0.9844	0.9735	0.9566	413.7699
stage1	4	0.0296	0.0137	0.9823	0.9704	0.9517	430.7479
stage1	5	0.0276	0.0147	0.9810	0.9684	0.9484	449.4992
stage1	6	0.0270	0.0115	0.9854	0.9754	0.9597	408.9824
stage1	7	0.0269	0.0117	0.9842	0.9736	0.9568	440.5142
stage1	8	0.0259	0.0114	0.9846	0.9741	0.9575	441.3940
stage1	9	0.0259	0.0143	0.9819	0.9700	0.9511	427.6779
stage1	10	0.0238	0.0110	0.9858	0.9760	0.9607	423.4209
stage2	11	0.0188	0.0070	0.9902	0.9835	0.9730	430.2548
stage2	12	0.0122	0.0049	0.9950	0.9915	0.9861	428.5968
stage2	13	0.0099	0.0048	0.9938	0.9896	0.9830	418.6639
stage2	14	0.0083	0.0044	0.9944	0.9905	0.9845	434.5803
stage2	15	0.0066	0.0034	0.9962	0.9935	0.9893	440.1876
stage2	16	0.0055	0.0038	0.9946	0.9909	0.9850	438.5227
stage2	17	0.0045	0.0032	0.9962	0.9934	0.9892	438.5588
stage2	18	0.0037	0.0033	0.9956	0.9925	0.9876	450.8087
stage2	19	0.0029	0.0034	0.9954	0.9922	0.9871	435.8280
stage2	20	0.0032	0.0034	0.9960	0.9931	0.9887	450.1781
stage2	21	0.0026	0.0032	0.9960	0.9931	0.9887	444.5331
stage2	22	0.0024	0.0031	0.9962	0.9935	0.9892	428.6202
stage2	23	0.0025	0.0036	0.9956	0.9925	0.9877	448.5754
stage2	24	0.0018	0.0031	0.9958	0.9928	0.9882	453.6805
stage2	25	0.0023	0.0037	0.9960	0.9931	0.9887	425.3830
stage2	26	0.0016	0.0033	0.9960	0.9931	0.9887	437.4319
stage2	27	0.0018	0.0032	0.9962	0.9934	0.9892	419.6439
stage2	28	0.0021	0.0035	0.9962	0.9935	0.9892	417.5124
stage2	29	0.0015	0.0034	0.9963	0.9938	0.9898	412.5772
stage2	30	0.0021	0.0031	0.9962	0.9934	0.9892	451.9737

Validation Forest F1 Over Epochs



Train Loss vs Validation Loss

EfficientNet-B2 Train / Validation Loss Curve

